METHODOLOGY ARTICLE

Assembly of chloroplast genome with short-read data: ways of optimization using *Heliantus niveus* and *Heliantus praecox* as a test case

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Abstract

Background: The chloroplast is an organelle that conducts photosynthesis in plant and algal cells. The genome of the chloroplast is a quadripartite structure characterized by one short single copy (SSC), one long single copy (LSC) and two inverted repeats (IR), although there are plants with atypical structures that for example lose the IR.
The assembly reference-based of chloroplast genome from short-read sequencing is a challenge that gives knowledge in many biology fields, like phylogenetics analysis, species identification, and genetic engineering. With this aim it's important to find the optimal conditions to perform an assembly that covers all the chloroplast genome, in this article I analyze the choice of the coverage and k-mer parameters, the use of different assembly tools, and finally the importance of the evolutionary distance of the reference genome plant respect to the genome plant for which we perform the assembly. The sequencing data come from *Heliantus niveus* and *Heliantus praecox.*

Results: The phase of the optimization of the assembly is done with short-read data from the chloroplast genome *Heliantus niveus sp. canescens* and as a reference chloroplast genome of *Heliantus debilis*. The assembly without optimization of the parameters results in a total length of 127 806 pb and 5 contigs, the assembly with optimization of coverage and k-mer results in a complete organelle genome with two contigs and total length 151 415 pb. The use of different assembly tool, in particular SOAP-denovo, results in an assembly of 129 175 pb as total lenght. Finally I have evaluated the use of a different species, related in terms of evolutionary distance to *H. niveus* and *H. debilis*. The result of the assembly comparison *H. niveus-H. debilis* and *H. praecox-H.debilis* consists in both the cases in a completed organelle and longest sequence composed by two IR and LSC.

Conclusion: My results shows that is possible to improve the assembly and select the best in term of statistics by finding the optimal value of coverage and k-mer and choosing the right assembly program. These factors have always to be tested because change in relation to many causes: characteristics of the sequencing reads, and the characteristics of the genome. Finally in a short-read reference based assembly, it's important to select the reference chloroplast genome of a related species, near in the phylogenetic tree, because more probably the coverage will be higher and the assembly performs better.

Keywords: Chloroplast genome, genome assembly, short-reads, k-mer, coverage, phylogenetics

Background

Plastids are a group of dynamic organelles in plants able to perform many functions. One prominent activity is photosynthesis, carried out by the green plastids, chloroplasts [4].
The chloroplast genome provides important information for phylogenetics, population-genetics and species identification [1-7], and it's also the focus of genetic engineering because it contains many genes involved in photosynthesis [1]. The chloroplast genome is a double-stranded DNA molecule of around 120kb – 160kb in size in most plants, encoding around 100 protein coding genes [9, 10]. It is usually circular, although some research suggests that it could be linear in certain developmental stages [11, 12]; it has an overall quadripartite structure that shows a high degree of conservation in land plants, despite frequent variations in IR/SC boundaries [3]. In particular the structure is composed by two copies of IR sequence, which varies in size from 6.0 kb (in *Pseudendoclonium akinetum*) to 76 kb (in *Pelargonium X hortorum*); these IR divide the genome into two distinct single copy (SC) regions (LSC and SSC regions) [3]. Accurate genome assembly is a key first step in the study of chloroplast genomes [5]. Sequence assembly is the process whereby one or more consensus sequences are reconstructed from hundreds to billions of

individual DNA sequence reads [6]. The programs to produce consensus sequences can be classified into two groups based on the algorithm they use: Overlap–Layout–Consensus (OLC) and De Bruijn Graph (DBG) [6].
Initial assemblies of the chloroplast genomes relied on Sanger sequencing, that has been largely replaced by short-read sequencing that produce a large amount of data for a lower cost, and has a high per-base accuracy. Most studies which assembled chloroplast genome from short-read data rely on assemblers designed for whole genome assembly, such as Abyss [10] and SOAPdenovo [11], that work with DBG algorithm. The short-read sequencing has some negative aspects like the difficulty to assemble the chloroplast genome into a single contig, because short-read data rarely contain sufficient long-range information to span an entire inverted repeat [5]. But can be useful and precise if the assembly is guided by a reference genome [12]. In this condition it's important to make the assumption that the chloroplast genome under study has not atypical structures, such as the *Chickpea*, which has lost one of the inverted repeat regions [8] or *Selaginella tamariscina* in which the repeats are in the same orientation instead of inverted [9].
In this study, I set out to determine the best approach to chloroplast genome assembly, starting from short reads of *Heliantus niveus* subsp. *canescens* for which no chloroplast genome is available, and with a reference-based approach (*H. debilis*). This species was of interest for potential breeding with *Heliantus annus*/common sunflower, because has a high temperature stress tolerance [19]. I performed the assembly also of the chloroplast genome of *Heliantus praecox,* a species near to both *H. niveus* and *H. debilis* in the phylogenetic tree. I followed a work pipeline during which I monitored the output after each stage. The goal is the optimization of the assembly; I tried to show how the choice of plants species, the improvement of the parameters of the assembly, and the use of different tools can act in the aim of optimization.

Material and methods

Illumina reads from *Heliantus niveus* (SRR8896549) and *Heliantus praecox* (SRR3492254), were downloaded from the SRA repository using the Prefech program from the SRA Toolkit. The SRA file format was then converted to Fastq format using the Fastq-dump program in the SRAToolkit. The reference genome was downloaded from the SRA repository as fasta file.
DNA extraction: genomic DNA was extracted from leaf tissue of *Helianthus niveus* using a modified CTAB protocol, DNA was sheared using a Covaris M220 ultrasonicator (Covaris, Woburn, Massachusetts, USA). End- repairing of the sheared DNA fragments was performed using the NEBNext End Repair Module (NEB, Ipswich, Massachusetts, USA). The reactions were purified and to reduce the representation of repetitive sequences, the enriched libraries were treated with a Duplex-Specific Nuclease (DSN; Evrogen, Moscow, Russia). The fragments were then further amplified using Kapa HiFi HotStart ReadyMix and added a six-bp index to the P7 adapter identical to that Illuminas TruSeq adapters. All libraries were sequenced at the Genome Quebc Innovation Center a HiSeq 4000 instrument (Illumina, San Diego, CA, USA). The library is pair-end with 2 x 150bp reads, 7.1G bases and 3.2 Gb.
For DNA extraction of *Heliantus praecox* there wasn't the procedure available. But the characteristics of the library are: 2x100pb reads, 4.2G bases and 2.7 Gb
Process and map reads: process raw reads with Fastq tools in particular fastq-mcf v.1.4.636 [13] that removes the adapter sequences from the fastq files. Selection of a percentage of reads with Seqtk (1.0-r31) [29]. Extract the chloroplast reads by attempting to align all reads to the reference genome (*Heliantus debilis*). This task is done with Bowtie2 v.2.3.4.1 [14], that indexes first the reference genome with an FM Index (based on the Burrows-Wheeler Transform or BWT) and after map the reads. These are sorted by position end indexed with Samtools v.1.7 [15]. I got the coverage per position with Bedtools v.2.29.2 [16], starting from the bam file and the indexed reference. With Samtools v.1.7. I extracted the mapped reads from the bam file.
Reads assembly: ABySS v.2.0.2 [17] is a de novo sequence assembler intended for short paired-end reads and large genomes. I describe the process of optimization of the assembly varying the parameters, and tools for the assembly in *Heliantus niveus*; although was made also for *Heliantus praecox* in the same way. *Heliantus praecox* is used later to discuss about the choice of evolutionary related species.
I perform different assemblies. In the first 15 assemblies I evaluated different coverages, by selecting different percentage of the processed reads at start. First with large ranges (5%, 20%, 50%, 70%, 90%) and after I focalized on the more interesting in terms of statistics (4 %, 16%, 17%) (Table 2). These assemblies were evaluated with a 63-mer. I did other 6 assembly with different k-mer size, with the same approach that I used for the evaluation of the coverage, so first with 31-mer, 61-mer, 81-mer and second in the region of interest (Table 3).
I did another assembly using SOAP-de novo v.2.04 [22] to see if the result could be improved.

Evaluation of different assembly performance: The candidate best assembly between all the different is based on these statistics (1) lowest number of scaffolds (2) fewest gaps (3) longest assembly length (4) lowest N50 index. I remapped the reads to the assembly to evaluate the coverage of the assembly.
Assembly on different species: Comparison *H. niveus* **-** *H. debilis* assembly with *H. praecox* **-** *H. debilis* assembly, through BlastN (2.3.0+) [28] alignment of the longest assembly sequence and the reference genome.

Results

In the mapping step at first I choose as a reference genome *Heliantus annus*/common sunflower (151 104 bp), but the mapping results have a lower coverage although using all the dataset, so I changed the reference genome to *H. debilis* (151 117 bp), species nearer to *H. niveus* in the phylogenetic tree.
In table 1 are reported the results of mapping and Abyss assembly using the whole dataset and a 63-mer size. For *H. niveus*, these conditions are not optimal because there are 25 contigs. Instead for *H. praecox* seems to be optimal with only 3 contigs and a total length of 175 673 that covers all the genome (coverage > 1).

Table 1. Summary raw, processed, mapped, assemblied chloroplast reads of *Heliantus niveus* and *Heliantus praecox*.
Red: Assembly to improve, Green: Optimal assembly

Subject genome	Reads raw	Reads processed	Mean length (pb)	Total bases raw (pb)	Total bases processed (pb)	Mapped reads	Assembly size (bp)
Heliantus niveus	47,183,022	46,958,444	150	7,077,453,330	7,031,045,952	257,753	127 806
Heliantus praecox	41, 944, 922	38,335,070	100	4,914,494,200	3,710,171,822	353,614	175 673

I continued with the improvement of the Abyss assembly of *H. niveus*, in particular in the evaluation of the coverage (table 2) I reached the optimal with percentage of initial processed reads equal to 17.1%, the total length is 127,063 and is represented by 5 contigs (Table 2).
In table 2 are reported also results from the variation of the coverage in *H. praecox*, the better assembly is confirmed at 100% of the dataset.

Table 2. Summary of the *Heliantus niveus* and *Heliantus praecox* chloroplast genome assembly, with coverage optimization. Yellow: Assembly to improve, Green: Optimal assembly

% Reads	Input processed reads	Subject genome	Mapped reads	Coverage (x)	Assembly size (bp)	Organelle completed[a]
4.3	2,018,930	*Heliantus niveus*	27,624	27	124,000	NO
16.0	7,570,992		39,099	38	126,978	NO
17.0	8,025,250		44,490	44	127,039	NO
17.1	8,050,488		44,641	44	127,063	NO*
17.8	8,379,573		44,783	44	127,039	NO
30	5,750,261	*Heliantus praecox*	106,300	68	126,885	NO
50	9,583,768		177,150	112	127,121	NO
80	15,334,028		282,912	180	151,053	NO
90	17,250,782		318,144	202	151,053	NO
100	**19,167,535**		**353,614**	**225**	**175,673**	**YES**

a The chloroplast genome assembly was considered complete when the difference in size compared to the reference genome was <10 nucleotides[6]
*the complete organelle is completed with the selection of the k-mers

I continued with the choice of the best k-parameter, the optimal final assembly was found with 77-mer, that results in two contigs and covers the entire chloroplast genome (total length 151 415 pb) (Table 3).

Table 3. Statistics of the *Heliantus niveus* chloroplast genome assembly, with k-mer optimization. Green: Optimal assembly

	k-31	**k-63**	**k-75**	**k-77**	**k-79**	**k-81**
Seq count	16	5	3	**2**	6	2
Total length	127 958	127 063	128 831	**151 415**	151 412	151 360
Longest seq	37591	80 524	84 534	**129 588**	129 585	129 533
Shortest seq	31	523	21 771	**21 827**	21 827	21 827
N50 len	2	22 541	22 526	**21 827**	21 827	21 827
N50 index	35 523	2	2	**2**	2	2

To asses with another approach the good quality of the Abyss assembly I remapped the reads to the assembly, the number of reads mapped results in 257,894 reads, that results in a 256x coverage. In the Rstudio v.3.5.3[30] graph, I saw that all the reads are spread uniformly and there aren't peaks related to the IR.
I tried to improve the assembly using another program, i.e. SOAP-denovo, the assembler gave the best assembly with 47% of the processed reads and 67-mer, with total length equal to 129 175 and longest sequence 48,356.
Finally, I have compared the result of the previous Abyss assembly with a new reference-based assembly of *H. praecox* that like *H. niveus* is near to *H. debilis* in the phylogenetic tree (Fig.1).

Figure 1. Phylogenetic tree for *Helianthus* [21]

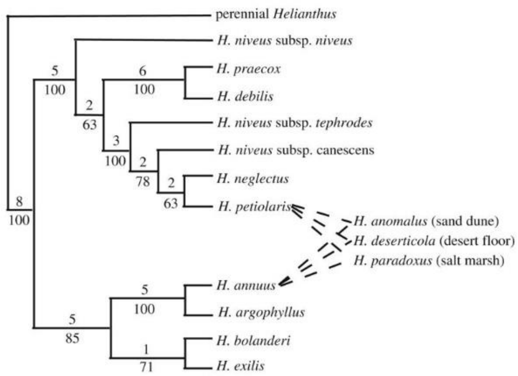

Table 4. Comparison assembly statistics, both with reference *H. debilis*

	Heliantus niveus	***Heliantus praecox***
Assembly statistics	**17.1%, 77-mer**	**100%, 63-mer**
Seq count	2	3
Total length	151 415	175 673
Longest sequence	129 588	129 232
Shortest sequence	21 827	21 826
N50	2	2
Coverage	1	1

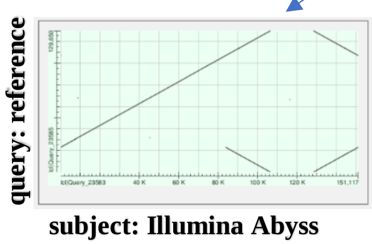

Figure 2. Blast. Alignment longest assembly seq. *H. niveus*-reference

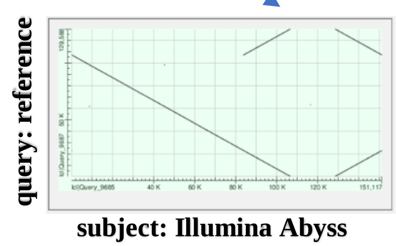

Figure 3. Blast. Alignment longest assembly seq. *H. praecox*-reference

The total length of the chloroplast assembly covers all the length of the chloroplast, for both the assemblies. The longest sequences are equal to ~129kb. The differences between the two are the percentage of initial dataset, and k-mer required.
I use the Basic Local Alignment Search Tool to perform an alignment between the reference chloroplast genome and the Abyss longest sequences of *H. niveus* and *praecox* (Figure2-3). The alignment is equal, in both the two assemblies it's possible to see the two inverted repeats and the longest sequence (LSC, ~129kb).

DISCUSSION
A factor influencing the assembly is the k-mer: both too short and too long k-mers are not optimal and lead to break of contigs. This is because long k-mers have more probability to not overlap of k-1 mer, because having a more quantity of bases, can also have a higher probability of sequencing error inside them so are more selective. This translates in gaps or no nodes in the Brujin graph. Short k-mers have an high variability and can overlap with many sequences, this can lead to regions more represented than others and so the formation of parts of the genome not covered; in terms of Brujin graph we will have many vertices, and an higher level of path ambiguity.
The other factor is the percentage of dataset taken, that influence the coverage of the mapped reads and so the assembly statistics. The optimal assembly was found with 17.1 % of the dataset, coverage 44 x; the increasing or decreasing of the percentage, leads to worst assembly, with an increasing of the number of contigs.
The choice of another program, i.e. SOAPdenovo, had not be successful in the optimizations, but it showed that the number of k-mer and the coverage required, change a lot between different assemblers although the two assembly programs work with the same approach (DBG).
With the remapping I saw that all the sequences have been assembled, the absence of IR peaks was maybe dued to the treatment in the preparation of the library (Duplex-Specific Nuclease) to reduce the representation of repetitive sequences.

Finally, I compared the assembly done with *H. niveus*-*H. debilis* and the assembly *H. praecox*-*H. debilis*. The quality and quantity of the initial reads were very similar (Table 1). Like *H. niveus*, *H. praecox* is near to *H. debilis* in the phylogenetic tree (Fig. 1), what we expect is a good performance of the assembly, that should be similar to *H. niveus.* The resulting assemblies are almost equal in goodness, the longest sequence is composed in both cases by the LSC and the two IR; the SSC is probably broken in another contigs, infact the longest sequence is only ~130 kb (total genome ~151 kb), probably the difference in ~21 kb is the SSC . The importance of the choice of the reference genome is tested also in the initial phase in which we initially use the *H. annus* as a reference (more distant in the phylogenetic tree) but we change because the mapped coverage was poor.
It was interesting denoting that the percentage of reads required for *H. praecox* was higher, this can be explained from the fact that the reads had an average length of 100 and not 150 like in *H. niveus,* read length smaller in size translates in less possible number of k-mer from them, this lead to a request of an higher amount of sequences. Infact from the formula, a sequence of length L will have L-k+1 possible k-mers. Also the choice of the k-mer changes, this shows that the choice of the parameters, i. e. coverage and k-mer, have to be chosen experimentally because is different when dealing with different genomes, although the genomes are evolutionary related.

CONCLUSION

The improvement of the short-read reference based assembly is based on first a careful choice of the parameters and the assembly program that have to be selected experimentally, through an evaluation of the statistics.
The impossibility to give a rule for the choice of the parameters is because the variation of these are related to different factors, in particular we see that some of them are characteristics of the reads (e. g. read length, protocols done in preparation of the library) and characteristics of the genome.
Last another important factor is the choice of the reference genome, as we see the choice of species near in the phylogenetic tree delivers similar results, because of the similarity of chloroplast genomes.

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